

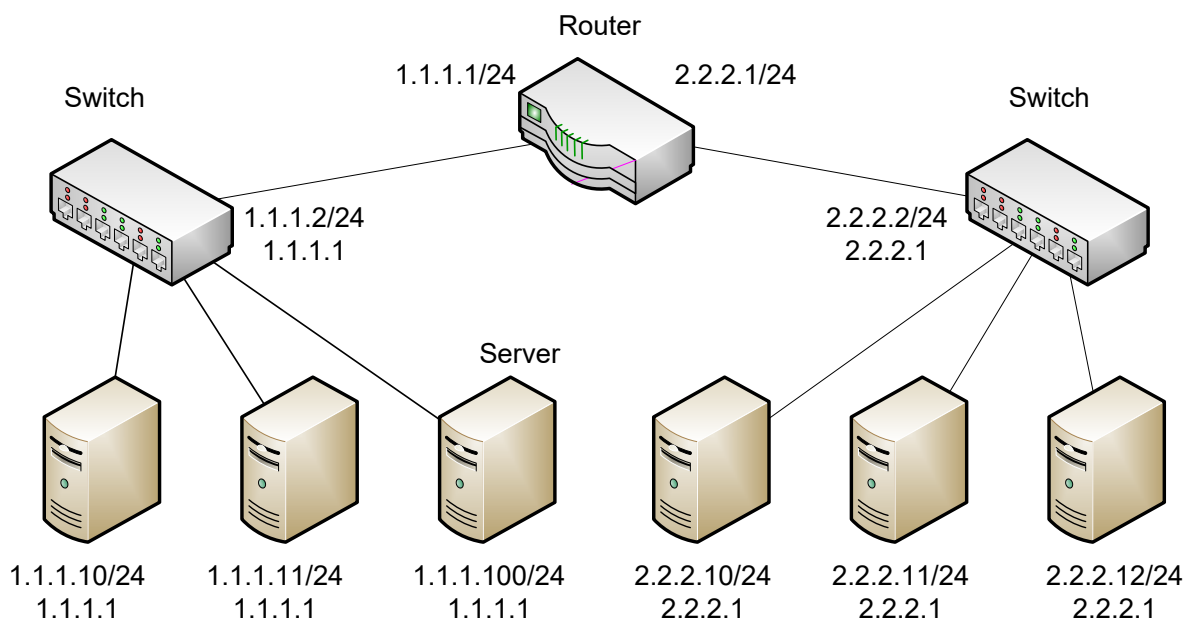
Lab 8

Basic Router Operation

Objective:

The purpose of this lab is for the student to understand basic router functionality. Students will also utilize IOS commands to explore router functionality and features.

Diagram:



Components:

6 – PC's with an Ethernet NIC

2 – Switches

1 – 2 Port Router

Procedure:

1. Connect all PC's and the server together and put their static IP addresses, subnet masks and gateways on them. Test connectivity by pinging each machine.
 - a. Can the machines within the 1.1.1.0/24 network ping any of the machines in the 2.2.2.0/24 network? Why?
No, they cannot ping each other because they are not on the same network.
2. Connect one PC to the router via a console cable.
3. Execute the Putty program.
4. What are the serial port settings needed to connect to the router?

Bits per second 9600

Data Bits 8

Parity None

Stop Bits 1

Flow Control None

5. What is the first prompt you see when you connect?
Router>
6. Get to the privileged exec prompt.
 - a. What command did you enter? enable
 - b. What does the prompt look like? Router#

7. View the running configuration.

- a. What command did you enter? show running-config
- b. What interfaces are available?
GigabitEthernet0/0/0 and GigabitEthernet0/0/1
- c. Are they currently turned on? GigabitEthernet0/0/0 is turned on, GigabitEthernet0/0/1 is turned off.

8. View the routing table.

- a. What command did you enter? show ip route

9. Change to global configuration mode.

- a. What command did you enter? configuration terminal

10. Change the router's hostname to Core-rtr.

- a. What command did you enter? hostname Core-rtr
- b. What does the prompt look like? Core-rtr(config)#

11. Select interface FastEthernet0/0.

- a. FastEthernet 0/0 says 10/100 on the front of the router.
- b. What command did you enter? interface GigabitEthernet0/0/0
- c. What does the prompt look like? Core-rtr(config-if)#

12. Put an IP address of 1.1.1.1 with a subnet mask of /24 on the interface.

- a. What command did you enter? ip address 1.1.1.1 255.255.255.0

13. Turn the interface on

- a. What command did you enter? no shutdown

14. View the routing table.

- a. What is displayed? One connected network and one local network

15. Select interface Ethernet0/0.

- a. Ethernet 0/0 says 10BT on the front of the router

16. Put an IP address of 2.2.2.1 with a subnet mask of /24 on the interface.

- a. What command did you enter? ip address 2.2.2.1 255.255.255.0

17. View the routing table.

- a. What is displayed? The same thing; one connected network and one local network
- b. Does the 2.2.2.0/24 network show up in the routing table? no
- c. Why? The interface is not on

18. Turn the interface on

- a. What command did you enter? no shutdown

19. View the routing table.

- a. What is displayed? Two connected networks and two local networks – both the 1.1.1.0 and 2.2.2.0 are up now.
- b. Does the 2.2.2.0/24 network show up in the routing table? Yes
- c. Why? Because the interface was turned on.

20. Can the router ping 1.1.1.10 and 2.2.2.10? Yes it can

21. Ping 2.2.2.10 from the 1.1.1.10 PC.

- a. Was the ping successful? Yes
- b. Were all four of the ping packets successful? Yes, surprisingly
- c. Why? When the router pinged the devices, it dropped the first ping, but the next four were correct – presumably the computers would have the same expected result: a delay in successful ping.

22. Change 1.1.1.10's gateway to 1.1.1.2

- a. Can the PC ping everyone in the 1.1.1.0/24 network? Yes
- b. Can the PC ping everyone in the 2.2.2.0/24 network? No
- c. Why? The computer was able to ping everyone in its LAN, but the ping failed to reach the router (its gateway) so it got dropped, and never connected to the 2 network.

23. Place IP addresses of 1.1.1.2/24 and 2.2.2.2/24 on the switches.

24. Can the switch with 1.1.1.2 on it ping the tftp server? no

25.Can the switch with 2.2.2.2 on it ping the tftp server? no

- a. Why? The default gateways have not been set

26.Put a gateway of 1.1.1.1 on the switch with an IP of 1.1.1.2

- a. What command did you enter? ip default-gateway 1.1.1.1
- b. Can the switch ping the server? yes
- c. Why? The gateway has been set

27.Put a gateway of 2.2.2.1 on the switch with an IP of 2.2.2.2

- a. What command did you enter? ip default-gateway 2.2.2.1
- b. Can the switch ping the server? yes
- c. Why? The gateway has been set

28.Turn on the TFTP service on the server.

29.Ping the server from the router

- a. What command did you enter? ping 1.1.1.100

30.Save the routers configuration to the router

- a. What command did you enter? copy running-config startup-config

31.Save the routers configuration to the TFTP server

a. What command did you enter? copy running-config tftp

32. Erase the routers configuration

a. What command did you enter? write erase

33. Draw the lab diagram below (PC's and hub connections) Label the IP addresses, and subnet masks of each device. Feel free to use packet tracer or Visio, but do not copy and paste the diagram at the beginning of the lab. Screen shot your diagram and paste it below:

